

Saurabh Nair

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EDUCATION

University of Toronto

Expct. Grad. Jun 2027

B.Sc Computer Science & B.A Economics (Double Major), CGPA: 3.73

- Dean's List Scholar (2022, 2023, 2024, 2025), University of Toronto Scholar Award (2022)

EXPERIENCE

Software Developer Intern

May 2025 – August 2026

BAYER Canada

Toronto, ON

- Owned end-to-end delivery of the DRL Redesign initiative on the Radimetrics dose management platform, shipping exam-level alert filtering, resulting in a *28% improvement* in alert filtering accuracy.
- Built an automated OS patch pipeline using Jenkins and AWS, replacing a manual release process with repeatable ISO packaging and provisioning, thus reducing release preparation time by *35%*.
- Led dependency upgrades across RHEL and Alma Linux environments for the 3.8 release, coordinating testing across multiple VMs, contributing to a *20% decline* in post-release defects.

Software Engineering Intern

Sep 2024 – May 2025

Kaytoons Inc.

San Jose, CA

- Built, deployed, and remotely maintained robust backend systems for a children's educational mobile app, Through proactive monitoring and code refactoring, improved *system uptime to 90%*, directly supporting investor demos.
- Integrated a PlayHT-powered voice cloning feature in Python, enabling dynamic generation of character dialogue thus enhancing user engagement and resulting in an *18% increase* in average session duration.

Teaching Assistant

Sep 2024 – Present

University of Toronto

Mississauga, ON

- Taught *CSC236: Theory of Computation*, and *ECO225: Data Tools for Economics*, *CSC263: Data Structures and Algorithms* introducing students to crucial early concepts in both Computer Science and Economics.
- Graded assignments, quizzes, and exams for both courses, provided detailed feedback, and held office hours to support over 150 students, resulting in improved engagement and higher course satisfaction scores.

Machine Learning Intern

Jun 2024 – Aug 2024

Emirates Steel Arkan

Abu Dhabi, UAE

- Developed, trained, and deployed predictive maintenance models leveraging time-series sensor data to anticipate equipment failures early thus reducing unplanned *downtime by 15%* saving hundreds of production hours annually.
- Engineered a comprehensive dashboard to monitor input consumption in real time. This solution gave plant managers actionable insights and resulted in an estimated *\$1.2M annual reduction* in raw material costs.
- Performed extensive feature engineering and hyperparameter tuning using, *improving model precision by 12%* and increasing the reliability for preventative maintenance activities across multiple production lines.

PROJECTS

Feynomenon – AI-Powered Learning Tutor | *Next.js, TailwindCSS, Node.js, MongoDB, Firebase, Gemini*

- Designed and developed an adaptive AI tutoring platform leveraging the Feynman Technique by using *Gemini API* for dynamic response generation. (*Winner at TMU Solution Hacks '25*)
- Built a fully responsive, mobile-friendly frontend with *Next.js* and *TailwindCSS*, integrated *Firebase* authentication, deployed via *Vercel* and *Railway*. Implemented MongoDB-backed analytics to store and retrieve session data.

Round1 – Multimodal AI Interviewer | *Next.js, TailwindCSS, Firebase, MediaPipe, Gemini, RoboFlow*

- Developed a interview simulator that conducts job-role-aware questioning, and applies a transparent LLM rubric for real-time scoring. Combined natural language understanding with behavioral metrics extracted via *MediaPipe*.
- Built an recruiter dashboard with interactive charts, transcripts, pass/fail thresholds, and engagement flags using *RoboFlow* image snapshots. and Integrated *Firebase* Hosting and Storage for secure, scalable deployment.

TECHNICAL SKILLS

Languages: Java, JavaScript/TypeScript, Python, C, R, PostgreSQL, MySQL, Stata, HTML/CSS, Bash/Shell

Frameworks: Spring Boot, React, Next.js, Node.js, Flask, Hibernate/JPA, TailwindCSS, Material-UI

Developer Tools: Git, Docker, Maven, Postman, Jenkins, AWS, GCP, JupyterHub, Linux, Firebase, Vercel, Railway

Libraries: Pandas, SciKit-Learn, OpenCV, RoboFlow, MediaPipe